Session-XIII: Actual groundwater situation: March last week

Program Schedule

S.No	Time	Content	Method	Materials	Expected outcomes
1	15 min.	Recap.	Large group discussion	Twelfth session contents, charts and markers.	Participants recollect the contents discussed in 12th session.
2	30 min.	Data analysis (Rainfall data).	Large group discussion	Charts and markers.	Participants estimate total rainfall received by consolidating the data from all the rain gauge stations in the HU.
3	20 min.	HESA - Estimation of actual groundwater situation by the end of May.	Small group discussion	Charts and markers.	Participants calculate actual groundwater situation by the end of May.
4	60 min.	Post- BBE.	Large group discussion	Charts and markers.	Participants participate in Post BBE and assess their knowledge on groundwater science.
5	20 min	Short Study- Soil testing.	Demonstration	Soil testing kit, soil sample, coloured paper strips to represent the quantity of the available nutrient.	Participants discuss the process of soil testing and the nutrients present in the soil.
6	30 min.	Gender work matrix.	Small group discussion	Charts and markers.	Participants discuss the need to empower women in decision making by involving them in FWS / Trainings.
7	10 min.	Homework: Collection of actual rainfall data from Nov to May.	Large group discussion	Charts and markers.	Participants will collect actual rainfall received (Nov to May).
8	10 min.	Evaluation.	Large group discussion	Charts and markers.	Participants share their learning.

Learning objectives

By the end of the session participants will be able to:

- Recap the previous session,
- Analyze LTE results using HESA,
- Analyze rainfall data,

- Estimate actual groundwater situation by the end of May, and
- Participate in post- BBE.

Process

1. Recap Time: 15 min.

Objective : Participants recap the contents discussed in the previous session.

Content : Session twelfth contents:

- LTE observation and data collection,
- PHM data analysis:
  - Number of functional borewells,
  - Average water levels, pumping hours, pumping days and discharge,
  - Perceived change (HESA), and
- Understand the importance of PHM assets maintenance.

Method : Small group discussion

Materials : Content of 12th FWS session, charts and markers.

Process

- Welcome participants and start the session with FWS pledge.
- In large group, encourage each participant to share a content that was discussed in previous session and the learning from that one.
- As you facilitate the recap, write participants responses on the chart paper.
- Discuss the contents and the learning from the previous session.

2. Data analysis (Rain fall data) Time: 30 min.

Objective : Participants estimate total rainfall received by consolidating the data from all the rain gauge stations in the HU.

Content : Estimation of total rainfall in the HU.

Method : Large group discussion

Materials : Charts and markers.

Process

- Encourage the participants to fill in the rainfall data on the chart.
- Remind the process of taking averages discussed in 5th session and encourage the participants to calculate the average rainfall of all the rain gauge stations of the HU.
- Emphasize that we may not receive uniform rainfall throughout our Hydrological Unit. It is better to average the rainfall of all the rain gauge stations in the HU month wise.
- If necessary, once again conduct the following experiment to understand the process of average for each month.
  - Take four empty one litre bottles and name each bottle with one rain gauge station and the month (June).
  - Fill each bottle with water according to the rainfall received by that rain gauge station for June month.
  - Transfer the water from four bottles into a big tumbler.
  - Again fill the bottles equally using the water in tumbler.

- Explain that each bottle represents the average rainfall of the Hydrological Unit for the month of June.
- Encourage to make averages for the other months (Nov, Dec, Jan, Feb, Mar, Apr and May).
- Encourage participants to state the steps in estimation of recharge using rain fall data.
- Appreciate the participants and correct the information if required.
- Estimate the actual recharge and share with the participants.

3. Estimation of actual groundwater situation by the end of May Time: 20 min

Objective : Participants estimate the actual groundwater situation by the end of May.

Content : Actual groundwater situation by the end of May.

Method : Large group discussion

Materials : Charts, markers, and calculators.

Process

- Ask the participants, how do we estimate groundwater balance based on recharge and draft?
- Remind them of the draft during Kharif and Rabi.
- Encourage one of the participants to explain the process.
- Total draft = Kharif draft + Rabi actual draft.
- Appreciate the participant and share the actual balance of groundwater by the end May.
- Actual groundwater balance by the end of May.

  = Recharge through actual rainfall – Actual draft.

  = Surplus / Deficit.
- Discuss increase or decrease in water balance and the encourage participants to share the reasons for change.
- Sensitize participants on adoption of water saving techniques and water use efficiency methods.

4. Post-BBE Time: 60 min**.**

Objective : Participants participate in Ballot Box Exercise.
Participants answer based on their knowledge.

Content : Ballot Box Exercise.

Method : Demonstration

Materials : Charts, markers, card boards, nails, long sticks, knife, thread, ballot boxes, and whistle.

Process

- Remind participants about the BBE conducted in the first session and discuss with them the purpose of conducting the exercise again.

Tips to facilitator

- The purpose of post-BBE is to know the increased knowledge and skills learnt by FWS participants.
- The present score will be compared with pre-BBE score, and
- Based on the increase in the score graduation certificates will be issued to eligible farmer participants.

- Assign a registration number to each participant.
- Give each participant 30 ballot slips (number of slips should be equal to the number of questions) bearing his/her registration number.
- Ask each participant to lineup in front of one cardboard station (i.e. one participant in front of one cardboard station).
- Each cardboard station has a question and three options (yes, no, no idea).
- Beneath the question, each option has a cover.
- Ask the participants to read the question, choose the appropriate option and drop the ballot into that cover.
- On blowing the whistle, participants move to the next station to read the question and cast their votes for the question.
- Participants exercise their ballot at each station.

Tips to facilitator

- Help participants distinguish between a cardboard station, a ballot, and a ballot Box.
- Before the conduct of the exercise, explain all the instructions to the participants.
- Organizing a mock demonstration before the actual conduct of the exercise will help participants.
- Assist illiterate or semi-literate farmers by reading the question and options.
- Only one participant should stand near one cardboard station.
- Do not allow two or more persons to crowd near one cardboard station.

Analysis of BBE Results

- First take out chits from 'No Idea' pocket, count them and record on a chart.
- After this, go to the pocket with least number of chits, count them and record on a chart.
- Finally record the third one by deducting the sum of the two from the total participants (There is no need to count the chits in the third pocket).
- Categorize the questions based on similarity.
- Calculate cumulative right/ wrong/ no idea question-wise and individual farmer-wise.
- Calculate percentages.

5. Short Study- Soil testing Time: 20 min.

Objective: Participants discuss the process of soil testing and the nutrients present in the soil.

Rationale: Soil testing is very important in crop production. Growing high yielding varieties and indiscriminate use of chemical fertilizers results in decreased productivity. It is essential to know the nutrients available in the soil by testing and the application of the nutrients based on the results.

Materials: Soil testing kit, soil sample, coloured paper strips to represent the quantity of the available nutrient.

Usage

Determining the quantity of nitrogen available in the soil

- Take one gram of soil sample into the beaker.

- Add 10 millilitres of nitrogen-1 solution to the beaker.
- Cap the beaker tightly and shake the beaker horizontally for 3 minutes. Then keep it on a stand for 3 minutes and again shake for 2 minutes.
- Pass the solution through a filter paper.
- Take 5 ml. of filtered solution to another beaker and add a pinch of nitrogen -2 powder using a spoon and shake for 2 minutes.
- Keep the beaker on a stand for 10 minutes.
- Compare the colour of the solution with standardized colour strips of nitrogen to know the nitrogen level of the soil.

Determining the quantity of phosphorous available in the soil

- Take 2 grams of soil sample into the beaker.
- Add a pinch of charcoal powder to soil in the beaker.
- Add 20 ml. of phosphorus solution -1 to the beaker.
- Cap the beaker tightly and shake the beaker in horizontal position for 3 minutes. Then keep it on a stand for 3 minutes and again shake for 2 minutes.
- Pass the solution through a filter paper.
- Take 5 ml. of filtered solution to another beaker and add 3 to 4 drops of phosphorus solution -2 using a dropper and shake for 5 minutes in vertical position.
- Again add 4 ml. of phosphorus solution -3 and add phosperous powder -4. Shake the beaker well.
- Keep the beaker on a stand for 10 minutes.
- Compare the colour of the solution with standardized colour strips of phosphorus to know the phosphorus level of the soil.

Determining the quantity of Potash available in the soil

- Take 2 grams of soil sample into the beaker.
- Add 10 ml. of Potash solution -1 to the beaker.
- Cap the beaker tightly and shake the beaker in horizontal position for 2 minutes.
- Pass the solution through a filter paper.
- Take 5 ml. of filtered solution to another beaker and add 3 to 4 drops of potash solution -2 using a dropper and shake well.
- Keep the beaker on the stand for 2 minutes.
- Compare the colour of the solution with standardized colour strips of potash to know the potash level of the soil.

6. Short Study- Gender work matrix

Time: 30 min.

Introduction:

People tend to not recognize the important role of women, who are actively participating in agriculture as well as in domestic work. Therefore, to learn about the participation of women in agriculture, this study has been conducted.

Objective:

By the end of the exercise, participants will able to:

1. Discuss the involvement of both women and men in bhindi cultivation.
2. Discuss the need to empower women in decision making by involving them in FWS / Trainings.

Process

Ask the following questions in the large group

a. List out the activities in which women are involved in agriculture practices?

b. Why women involvement is needed in agriculture?

- Encourage the farmers to list all the agricultural activities in which women are involved. After consensus, list them on the board and summarize.

- Divide the participants into small groups and ask them to work on the following questions in their respective groups:
    - What is the time spent by men and women on each activity in agricultural practices (matrix form)?
    - How much time is required for each activity for men's and women's involvement in domestic work (matrix form)?

Given below is a sample format. Adapt this to your context.

Gender Work Matrix in crop Production:

S.no	Operation	No of days (8 her/ day)		Decision	
		Women	Men	Women	Men
1	Ploughing / Harrowing				
2	Cleaning field				
3	FYM - Transport - Application				
4	Seed selection				
5	Seed purchase				
6	Ridge & furrow preparation				
7	Sowing				
8	Irrigation				
9	Weeding				
10	Fertilizer/ purchase				
11	Fertilizer application				
12	Pesticide selection				

13	Pesticide purchase				
14	Pesticide application				
15	Harvest				
16	Marketing transport				
17	Marketing				
18	Money transaction				

7. **Homework** Time: 10 min.

- Request participants to discuss and get the actual rain fall data from November to May. Ask them to take the assistance of respective GMCs.
- Ask them to collect data on secondary water bodies (Total area of the tank or channel, water column).

8. **Evaluation** Time: 10min.

Ask the following questions to evaluate the day's learning. Pose the questions to the large group:

- What are the parameters required to calculate the total recharge?
- Why we need to estimate recharge again using rain fall data received during Nov to May?
- What is the purpose of the post- BBE?